

EC504 Fall 2025 Homework 1 Solution

Start reading Chapters 1, 2, 3 and 4 in CLRS . They give a very readable introduction to Algorithms.. Also glance at Appendix A for math tricks which we will use from time to time.

NOTE: THESE SOLUTIONS OFTEN HAVE FAR MORE DETAILS THAN REQUIRED. The purpose is to provide instructive to solve in-class questions and in-class discussions.

1. (40 pts) In CLRS do Exercise 3.1-4 on page 53, Problem 3.2-7 on page 60, Problem 3-2 on page 61 , Problem 4.3-9 on page 88.

Missing Solution to Exercise 3.1-4

Is $2^{(n+1)} = O(2^n)$ **TRUE** since $2^{(n+1)} = 2 * 2^n = \text{constant} * 2^n$

is $2^{(2n)} = O(2^n)$ **FALSE** since $2^{(2n)} = 2^n * 2^n$ there is not constant multiply: note limit $2^{(2n)}/2^n = 2^n \rightarrow \text{infinity}$

Problem 1a : Problem 3.2-7

Prove by induction that the i^{th} Fibonacci number satisfies the equality:

$$F_i = \frac{\phi^i - \hat{\phi}^i}{\sqrt{5}}$$

Recall:

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \hat{\phi} = \frac{1 - \sqrt{5}}{2}$$

Base case: We know that $F(0) = 0$ and $F(1) = 1$. We will show that the claim holds for both $i = 0$ and $i = 1$:

$$F_0 = \left(\frac{\phi^i - \hat{\phi}^i}{\sqrt{5}} \right) \Big|_{i=0} = \frac{1 - 1}{\sqrt{5}}$$
$$\therefore F_0 = 0 \quad \square$$

$$F_1 = \left(\frac{\phi^i - \hat{\phi}^i}{\sqrt{5}} \right) \Big|_{i=1} = \frac{\phi - \hat{\phi}}{\sqrt{5}} = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right) - \left(\frac{1 - \sqrt{5}}{2} \right) \right] = \frac{1}{\sqrt{5}} \left[\frac{1 + \sqrt{5} - 1 + \sqrt{5}}{2} \right] = \frac{1}{\sqrt{5}} \left(\frac{2\sqrt{5}}{2} \right)$$
$$\therefore F_1 = 1 \quad \square$$

Induction step: Assume the claim holds for i and $i - 1$ - our induction hypothesis is as follows:

$$F_i = \frac{\phi^i - \hat{\phi}^i}{\sqrt{5}}, \quad F_{i-1} = \frac{\phi^{i-1} - \hat{\phi}^{i-1}}{\sqrt{5}}$$

Show that if the claim holds for i and $i - 1$, the claim must hold for $i + 1$, namely:

$$F_{i+1} = \frac{\phi^{i+1} - \hat{\phi}^{i+1}}{\sqrt{5}}$$

By definition of the Fibonacci numbers:

$$F_{i+1} = F_i + F_{i-1}$$

Using the induction hypothesis:

$$F_{i+1} = \frac{\phi^i - \hat{\phi}^i}{\sqrt{5}} + \frac{\phi^{i-1} - \hat{\phi}^{i-1}}{\sqrt{5}}$$
$$F_{i+1} = \frac{1}{\sqrt{5}} (\phi^i + \phi^{i-1}) - \frac{1}{\sqrt{5}} (\hat{\phi}^i + \hat{\phi}^{i-1})$$

Using $\phi = \frac{1 + \sqrt{5}}{2}$:

$$\phi^i + \phi^{i-1} = \left(\frac{1 + \sqrt{5}}{2} \right)^i + \left(\frac{1 + \sqrt{5}}{2} \right)^{i-1}$$
$$\phi^i + \phi^{i-1} = \left(\frac{1 + \sqrt{5}}{2} \right)^i \left[1 + \frac{2}{1 + \sqrt{5}} \right]$$

$$\begin{aligned}
\phi^i + \phi^{i-1} &= \phi^i \left(\frac{1 + \sqrt{5} + 2}{1 + \sqrt{5}} \right) = \phi^i \left(\frac{3 + \sqrt{5}}{1 + \sqrt{5}} \right) \\
\phi^i + \phi^{i-1} &= \phi^i \left(\frac{3 + \sqrt{5}}{1 + \sqrt{5}} \right) \left(\frac{1 - \sqrt{5}}{1 - \sqrt{5}} \right) = \phi^i \left(\frac{3 - 3\sqrt{5} + \sqrt{5} - 5}{1 - \sqrt{5} + \sqrt{5} - 5} \right) = \phi^i \left(\frac{-2 - 2\sqrt{5}}{-4} \right) = \phi^i \left(\frac{-2(1 + \sqrt{5})}{-4} \right) \\
\phi^i + \phi^{i-1} &= \phi^i \times \left(\frac{1 + \sqrt{5}}{2} \right) = \phi^i \times \phi \\
\therefore \phi^i + \phi^{i-1} &= \phi^{i+1}
\end{aligned}$$

Similarly, using $\hat{\phi} = \frac{1-\sqrt{5}}{2}$:

$$\begin{aligned}
\hat{\phi}^i + \hat{\phi}^{i-1} &= \left(\frac{1 - \sqrt{5}}{2} \right)^i + \left(\frac{1 - \sqrt{5}}{2} \right)^{i-1} \\
\hat{\phi}^i + \hat{\phi}^{i-1} &= \left(\frac{1 - \sqrt{5}}{2} \right)^i \left[1 + \frac{2}{1 - \sqrt{5}} \right] \\
\hat{\phi}^i + \hat{\phi}^{i-1} &= \hat{\phi}^i \left(\frac{1 - \sqrt{5} + 2}{1 - \sqrt{5}} \right) = \hat{\phi}^i \left(\frac{3 - \sqrt{5}}{1 - \sqrt{5}} \right) \\
\hat{\phi}^i + \hat{\phi}^{i-1} &= \hat{\phi}^i \left(\frac{3 - \sqrt{5}}{1 - \sqrt{5}} \right) \left(\frac{1 + \sqrt{5}}{1 + \sqrt{5}} \right) = \hat{\phi}^i \left(\frac{3 + 3\sqrt{5} - \sqrt{5} - 5}{1 + \sqrt{5} - \sqrt{5} - 5} \right) = \hat{\phi}^i \left(\frac{-2 + 2\sqrt{5}}{-4} \right) = \hat{\phi}^i \left(\frac{-2(1 - \sqrt{5})}{-4} \right) \\
\hat{\phi}^i + \hat{\phi}^{i-1} &= \hat{\phi}^i \times \left(\frac{1 - \sqrt{5}}{2} \right) = \hat{\phi}^i \times \hat{\phi} \\
\therefore \hat{\phi}^i + \hat{\phi}^{i-1} &= \hat{\phi}^{i+1}
\end{aligned}$$

Putting it all together:

$$\begin{aligned}
F_{i+1} &= \frac{1}{\sqrt{5}} (\phi^i + \phi^{i-1}) - \frac{1}{\sqrt{5}} (\hat{\phi}^i + \hat{\phi}^{i-1}) = \frac{1}{\sqrt{5}} \phi^{i+1} - \frac{1}{\sqrt{5}} \hat{\phi}^{i+1} \\
\therefore F_{i+1} &= \frac{\phi^{i+1} - \hat{\phi}^{i+1}}{\sqrt{5}} \quad \square
\end{aligned}$$

Therefore, by induction, we have shown that the i^{TH} Fibonacci number can be expressed as $F_i = \frac{\phi^i - \hat{\phi}^i}{\sqrt{5}}$.

Problem 1a : Problem 3-2

Indicate, for each pair of expressions (A, B) in the table below whether A is $\{O, o, \Omega, \omega, \Theta\}$ of B . Assume that $k \geq 1$, $\epsilon > 0$, and $c > 1$.

A	B	O	o	Ω	ω	Θ
$\lg^k n$	n^ϵ	✓	✓	×	×	×
n^k	c^n	✓	✓	×	×	×
$\sqrt{n}$	$n^{\sin n}$	×	×	×	×	×
2^n	$2^{n/2}$	×	×	✓	✓	×
$n^{\lg c}$	$c^{\lg n}$	✓	×	✓	×	✓
$\lg n!$	$\lg n^n$	✓	×	✓	×	✓

Recalling the graphical relationship between the various asymptotic notations:

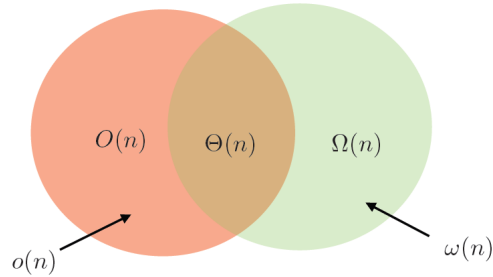


Figure 1: Relationship between asymptotic notations

We can write the formal definitions:

$$f(n) \in O((g(n))) \iff \exists c > 0, n_0 \text{ s.t. } f(n) \leq cg(n) \forall n \geq n_0$$

$$f(n) \in o((g(n))) \iff \forall c > 0, n_0 \text{ s.t. } f(n) < cg(n) \forall n \geq n_0$$

$$f(n) \in \Omega((g(n))) \iff \exists c > 0, n_0 \text{ s.t. } f(n) \geq cg(n) \forall n \geq n_0$$

$$f(n) \in \omega((g(n))) \iff \forall c > 0, n_0 \text{ s.t. } f(n) > cg(n) \forall n \geq n_0$$

We can write the limit definitions:

$$f(n) \in O((g(n))) \iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} < \infty$$

$$f(n) \in o((g(n))) \iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$$

$$f(n) \in \Omega((g(n))) \iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} > 0$$

$$f(n) \in \omega((g(n))) \iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$$

$$f(n) \in \Theta(g(n)) \iff f(n) \in O(g(n)) \wedge f(n) \in \Omega(g(n))$$

It is important to note that if $f(n) \in o(g(n))$, then $f(n) \in O(g(n))$; this follows from the limit definitions above. Similarly, if $f(n) \in \omega(g(n))$, then $f(n) \in \Omega(g(n))$.

$\lg^k n$ vs. n^ϵ :

Let $n = 2^m$, $m \geq 0$. Then we can express the limit as:

$$\lim_{n \rightarrow \infty} \frac{\lg^k n}{n^\epsilon} = \lim_{n \rightarrow \infty} \frac{(\lg n)^k}{n^\epsilon} = \lim_{m \rightarrow \infty} \frac{(\lg 2^m)^k}{(2^m)^\epsilon}$$

Then:

$$\lim_{m \rightarrow \infty} \frac{(m \lg 2)^k}{2^{m\epsilon}} = \lim_{m \rightarrow \infty} \frac{m^k}{2^{m\epsilon}} = \lim_{m \rightarrow \infty} \frac{m^k}{(2^\epsilon)^m}$$

Let $\alpha = 2^\epsilon$; since $\epsilon > 0$, $\alpha = 2^\epsilon > 1$:

$$\lim_{m \rightarrow \infty} \frac{m^k}{\alpha^m} = 0$$

Thus, $\lg^k n \in o(n^\epsilon)$; since $\lg^k n \in o(n^\epsilon)$, we also have that $\lg^k n \in O(n^\epsilon)$.

n^k vs. c^n :

Taking the limit:

$$\lim_{n \rightarrow \infty} \frac{n^k}{c^n}$$

Since $c > 1$, we note the same asymptotic behavior as the previous problem:

$$\lim_{n \rightarrow \infty} \frac{n^k}{c^n} = 0$$

Thus, $n^k \in o(c^n)$; since $n^k \in o(c^n)$, we also have that $n^k \in O(c^n)$.

$\sqrt{n}$ vs. $n^{\sin n}$:

We cannot make any asymptotic comparison between the two functions. Recall that $\sqrt{n}$ is a strictly increasing function. Thus, if $\sqrt{n} \in O(g(n))$, $\exists c > 0$ such that $g(n)$ is also strictly increasing for $n \geq n_0$. The function $g(n) = n^{\sin n}$ cannot satisfy this requirement; Let $N = \{n : n \geq n_0, \sin n = -1\}$ ¹. Since $\sin n$ is a periodic function, there are an infinite number of points in N . However, for all points in N :

$$n^{\sin n} = n^{-1} = \frac{1}{n} \quad \forall n \in N$$

However, $\sqrt{n} \geq \frac{1}{n} \quad \forall n \geq 1$; we can always find an infinite number of points where $n \geq n_0$ such that $f(n) = \sqrt{n} > g(n) \quad \forall n \in N$, violating the bound. We cannot use the constant c to help reverse the inequality, as we would need c to change for an infinite number of points; thus we would never be able to find a constant c to always satisfy the inequality $f(n) \leq cg(n) \quad \forall n \geq n_0$. Therefore, $\sqrt{n} \notin O(n^{\sin n})$. Since $\sqrt{n} \notin O(n^{\sin n})$, we can immediately conclude that $\sqrt{n} \notin o(n^{\sin n})$ and $\sqrt{n} \notin \Theta(n^{\sin n})$.

We can repeat the argument, except for a lower bound - the argument proceeds in the exact same way, and we can conclude that $\sqrt{n} \notin \Omega(n^{\sin n})$. Since $\sqrt{n} \notin \Omega(n^{\sin n})$, we can immediately conclude that $\sqrt{n} \notin \omega(n^{\sin n})$.

2^n vs. $2^{n/2}$

Computing the limit, we see:

$$\lim_{n \rightarrow \infty} \frac{2^n}{2^{n/2}} = \lim_{n \rightarrow \infty} \frac{2^n}{(2^{1/2})^n} = \lim_{n \rightarrow \infty} \left(\frac{2}{\sqrt{2}} \right)^n = \lim_{n \rightarrow \infty} (\sqrt{2})^n = \infty$$

¹I am implicitly extending n to the reals here - if n is an integer then the equality will not hold, but the general structure of the argument still holds. It's conceptually easier to argue from the reals.

Thus, $2^n \in \omega(2^{n/2})$; since $2^n \in \omega(2^{n/2})$, we also have that $2^n \in \Omega(2^{n/2})$.

$n^{\lg c}$ vs. $c^{\lg n}$:

Let $\lg c = q$; then $c = 2^q$. Let $n = 2^x$. Then we can express $n^{\lg c}$ as:

$$n^{\lg c} = (2^x)^{\lg c} = \left(2^{\lg c}\right)^x = c^x$$

But since $n = 2^x$, we have $x = \lg n$. Thus:

$$n^{\lg c} = c^x = c^{\lg n}$$

$$\therefore n^{\lg c} = c^{\lg n}$$

Since $n^{\lg c} = c^{\lg n}$, we have $f(n) = g(n)$; thus, by the reflexivity property, $f(n) \in O(f(n))$, $f(n) \in \Omega(f(n))$, $f(n) \in \Theta(f(n))$.

Thus, $n^{\lg c} \in O(c^{\lg n})$, $n^{\lg c} \in \Omega(c^{\lg n})$, and $n^{\lg c} \in \Theta(c^{\lg n})$.

$\lg n!$ vs. $\lg n^n$:

Using Stirling's approximation, we can express $n!$ as:

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + \Theta\left(\frac{1}{n}\right)\right)$$

Thus $\lg n!$ can be expressed as:

$$\lg n! = \lg \left[\sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + \Theta\left(\frac{1}{n}\right)\right) \right] = \lg \left[\sqrt{2\pi n} \left(\frac{n}{e}\right)^n \right] + \lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]$$

$$\lg n! = \lg \sqrt{2\pi n} + \lg \left(\frac{n}{e}\right)^n + \lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]$$

$$\lg n! = \lg (2\pi n)^{1/2} + n \lg \left(\frac{n}{e}\right) + \lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]$$

$$\lg n! = \frac{1}{2} \lg (2\pi n) + n (\lg n - \lg e) + \lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]$$

$$\lg n! = \frac{1}{2} (\lg 2 + \lg \pi + \lg n) + n \lg n - n \lg e + \lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]$$

$$\lg n! = \frac{1}{2} + \frac{\lg \pi}{2} + \frac{1}{2} \lg n + n \lg n - n \lg e + \lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]$$

$$\therefore \lg n! = n \lg n + \frac{1}{2} \lg n - (\lg e)n + \left(\frac{1 + \lg \pi}{2}\right) + \lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]$$

We can easily express $\lg n^n$ as $n \lg n$. Taking the limit:

$$\lim_{n \rightarrow \infty} \frac{\lg n!}{\lg n^n} = \lim_{n \rightarrow \infty} \frac{n \lg n + \frac{1}{2} \lg n - (\lg e)n + \left(\frac{1 + \lg \pi}{2}\right) + \lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]}{n \lg n}$$

$$\lim_{n \rightarrow \infty} \frac{\lg n!}{\lg n^n} = \lim_{n \rightarrow \infty} \frac{n \lg n}{n \lg n} + \frac{1/2 \lg n}{n \lg n} - \frac{(\lg e)n}{n \lg n} + \frac{(1 + \lg \pi)/2}{n \lg n} + \frac{\lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]}{n \lg n}$$

$$\lim_{n \rightarrow \infty} \frac{\lg n!}{\lg n^n} = 1 + \frac{1}{2n} - \frac{\lg e}{\ln n} + \frac{(1 + \lg \pi)/2}{n \lg n} + \frac{\lg \left[\left(1 + \Theta\left(\frac{1}{n}\right)\right) \right]}{n \lg n}$$

$$\lim_{n \rightarrow \infty} \frac{\lg n!}{\lg n^n} = 1 + \lim_{n \rightarrow \infty} \frac{\lg \left[\left(1 + \Theta \left(\frac{1}{n} \right) \right) \right]}{n \lg n}$$

By definition, for sufficiently large n , if $f(n) = \Theta \left(\frac{1}{n} \right)$, then $\exists c_2 > c_1 > 0$, s.t. $\frac{c_1}{n} \leq f(n) \leq \frac{c_2}{n}$. Thus:

$$\lg \left[1 + \Theta \left(\frac{1}{n} \right) \right] \sim \lg \left(1 + \frac{c}{n} \right) = \lg \left(\frac{n+c}{n} \right) = \lg (n+c) - \lg n$$

Using this result in the limit:

$$\lim_{n \rightarrow \infty} \frac{\lg n!}{\lg n^n} = 1 + \lim_{n \rightarrow \infty} \frac{\lg \left[\left(1 + \Theta \left(\frac{1}{n} \right) \right) \right]}{n \lg n} = 1 + \lim_{n \rightarrow \infty} \frac{\lg (n+c) - \lg n}{n \lg n}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{\lg n!}{\lg n^n} = 1$$

Thus, we can conclude that $\lg n! \in O(\lg n^n)$, $\lg n! \in \Omega(\lg n^n)$, and thus $\lg n! \in \Theta(\lg n^n)$.

Problem 1a : Problem 4.3-9

Solve the recurrence $T(n) = 3T(\sqrt{n}) + \lg n$ by making a change of variables.

For simplicity, assume $\log n = \lg n$; let $n = 2^m$. Then:

$$T(n) = 3T(\sqrt{n}) + \lg n \iff T(2^m) = 3T(\sqrt{2^m}) + \lg 2^m$$

$$T(2^m) = 3T(2^{m/2}) + m \lg 2$$

$$T(2^m) = 3T(2^{m/2}) + m$$

Let $\tilde{T}(m) = T(2^m)$; then we can write:

$$\tilde{T}(m) = 3\tilde{T}\left(\frac{m}{2}\right) + m$$

This recurrence relation fits the form of the master equation, with $a = 3$, $b = 2$, $f(m) = m$, and $k = 1$. Computing $\gamma = \frac{\ln a}{\ln b} = \log_2 3 > 1$, we see that $\gamma > k$; thus, we can immediately write the solution to the equation:

$$\tilde{T}(m) = \Theta(m^\gamma)$$

Recalling that $n = 2^m$, we can reverse the change of variables and find an expression for $T(n)$:

$$\tilde{T}(m) \in \Theta(m^\gamma)$$

$$\tilde{T}(\lg n) \in \Theta((\lg n)^\gamma)$$

Since $\tilde{T}(m) = T(2^m)$:

$$T(2^{\lg n}) \in \Theta((\lg n)^\gamma)$$

$$\therefore T(n) \in \Theta((\lg n)^\gamma)$$

Thus, $T(n) \in \Theta(\lg^\gamma n)$ with $\gamma = \log_2 3 > 1$.

Problem 1b

Place the following functions in order from asymptotically smallest to largest using $f(n) \in O(g(n))$ notation:

$$n^2 + 3n \log n + 5, \quad n^2 + n^{-2}, \quad n^{n^2} + n!, \quad n^{\frac{1}{n}}, \quad n^{n^2-1}, \quad \ln n, \quad \ln \ln n, \quad 3^{\ln n}, \quad 2^n$$

$$(1+n)^n, \quad n^{1+\cos n}, \quad \sum_{k=1}^{\log n} \frac{n^2}{2^k}, \quad 1, \quad n^2 + 3n + 5, \quad \log n!, \quad \sum_{k=1}^n \frac{1}{k}, \quad \prod_{k=1}^n \left(1 - \frac{1}{k^2}\right), \quad \left(1 - \frac{1}{n}\right)^n$$

Function $f(n) \in O(g(n))$ if $\exists c > 0$ such that:

$$f(n) \leq cg(n) \quad \forall n \geq n_0$$

Equivalently, we can use the limit definition:

$$f(n) \in O(g(n)) \iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} < \infty$$

$f_1(n) = n^2 + 3n \log n + 5$:

$$\lim_{n \rightarrow \infty} \frac{f_1(n)}{n^2} = \lim_{n \rightarrow \infty} \left(\frac{n^2 + 3n \log n + 5}{n^2} \right) = \lim_{n \rightarrow \infty} \left(1 + \frac{3 \log n}{n} + \frac{5}{n^2} \right) = 1$$

$\therefore f_1(n) = n^2 + 3n \log n + 5 \in O(n^2)$

$f_2(n) = n^2 + n^{-2}$:

$$\lim_{n \rightarrow \infty} \frac{f_2(n)}{n^2} = \lim_{n \rightarrow \infty} \left(\frac{n^2 + n^{-2}}{n^2} \right) = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n^4} \right) = 1$$

$\therefore f_2(n) = n^2 + n^{-2} \in O(n^2)$

$f_3(n) = n^{n^2} + n!$:

$$\lim_{n \rightarrow \infty} \frac{f_3(n)}{n^{n^2}} = \lim_{n \rightarrow \infty} \frac{n^{n^2} + n!}{n^{n^2}} = \lim_{n \rightarrow \infty} \left(1 + \frac{n!}{n^{n^2}} \right) = 1 + \lim_{n \rightarrow \infty} \frac{n!}{n^{n^2}}$$

By definition:

$$n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$$

$$n! = \prod_{i=1}^n i < \prod_{i=1}^n n < n^n < n^{n^2}$$

Therefore:

$$\lim_{n \rightarrow \infty} \frac{n!}{n^{n^2}} = 0$$

$\therefore f_3(n) = n^{n^2} + n! \in O(n^{n^2})$

$f_4(n) = n^{\frac{1}{n}}$:

$$\lim_{n \rightarrow \infty} \frac{f_4(n)}{1} = \lim_{n \rightarrow \infty} \left(\frac{n^{\frac{1}{n}}}{1} \right) = \lim_{n \rightarrow \infty} n^{1/n}$$

Evaluating this limit, we arrive at the indeterminate form (∞^0) ; we can solve by transforming the limit as follows:

$$\lim_{n \rightarrow \infty} f(x)^{g(x)} = \exp \lim_{n \rightarrow \infty} \frac{\ln f(x)}{\left(\frac{1}{g(x)}\right)} = \exp \left(\lim_{n \rightarrow \infty} g(x) \ln f(x) \right)$$

Therefore, with $f(x) = n$, $g(x) = \frac{1}{n}$:

$$\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = \exp \left(\lim_{n \rightarrow \infty} \left(\frac{1}{n} \right) (\ln n) \right) = \exp \left(\lim_{n \rightarrow \infty} \left(\frac{\ln n}{n} \right) \right) = \exp 0 = 1$$

$$\therefore f_4(n) = n^{\frac{1}{n}} \in O(1)$$

$$f_5(n) = n^{n^2-1}:$$

$$\lim_{n \rightarrow \infty} \frac{f_5(n)}{n^{n^2}} = \lim_{n \rightarrow \infty} \frac{n^{n^2-1}}{n^{n^2}} = \lim_{n \rightarrow \infty} \left(\frac{1}{n^{n^2}} \times \frac{n^{n^2}}{n} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\therefore f_5(n) = n^{n^2-1} \in O(n^{n^2})$$

$$f_6(n) = \ln n:$$

$$\lim_{n \rightarrow \infty} \frac{f_6(n)}{\ln n} = \frac{\ln n}{\ln n} = 1$$

$$\therefore f_6(n) = \ln n \in O(\ln n)$$

$$f_7(n) = \ln \ln n:$$

$$\lim_{n \rightarrow \infty} \frac{f_7(n)}{\ln \ln n} = \frac{\ln \ln n}{\ln \ln n} = 1$$

$$\therefore f_7(n) = \ln \ln n \in O(\ln \ln n)$$

$$f_8(n) = 3^{\ln n}:$$

Let $\ln n = m$, and let $e^x = 3$, $x = \ln 3$; then:

$$3^{\ln n} = (e^x)^m = e^{xm} = (e^m)^x$$

$$3^{\ln n} = \left(e^{\ln n} \right)^x = n^x = n^{\ln 3} = n^\gamma, \gamma = \ln 3$$

$$\therefore f_8(n) = 3^{\ln n} \in O(n^\gamma), \gamma < 2$$

$$f_9(n) = 2^n:$$

$$\lim_{n \rightarrow \infty} \frac{f_9(n)}{2^n} = \lim_{n \rightarrow \infty} \frac{2^n}{2^n} = 1$$

$$\therefore f_9(n) = 2^n \in O(2^n)$$

$$f_{10}(n) = (1+n)^n:$$

$$\lim_{n \rightarrow \infty} \frac{f_{10}(n)}{n^n} = \lim_{n \rightarrow \infty} \frac{(1+n)^n}{n^n}$$

Using the binomial theorem to expand $(1+n)^n$:

$$(1+n)^n = \sum_{k=0}^n \binom{n}{k} (1)^{n-k} n^k = \sum_{k=0}^n \binom{n}{k} n^k = n^n + \sum_{k=0}^{n-1} \binom{n}{k} n^k = n^n + \sum_{k=0}^{n-1} c_k n^k$$

$$\lim_{n \rightarrow \infty} \frac{(1+n)^n}{n^n} = \lim_{n \rightarrow \infty} \left(\frac{n^n + \sum_{k=0}^{n-1} c_k n^k}{n^n} \right) = \lim_{n \rightarrow \infty} \left(1 + \sum_{k=0}^{n-1} c_k \frac{n^k}{n^n} \right) = 1$$

$$\therefore f_{10}(n) = (1+n)^n \in O(n^n)$$

$$f_{11}(n) = n^{1+\cos n};$$

Use the definition:

$$f(n) \in O(g(n)) \iff \exists c < 0 \text{ s.t. } f(n) \leq cg(n) \forall n \geq n_0$$

By definition, $\cos n \in [-1, 1]$; thus, $(1 + \cos n) \in [0, 2]$. Since $(1 + \cos n) \leq 2 \forall n \geq 1$, we can bound $n^{1+\cos n}$ accordingly: $n^{1+\cos n} \leq n^2 \forall n \geq 1$.

$$\therefore f_{11}(n) = n^{1+\cos n} \in O(n^2)$$

$$f_{12}(n) = \sum_{k=1}^{\log n} \frac{n^2}{2^k};$$

Let $b = \log n$; then the sum can be expressed as:

$$f_{12}(n) = \sum_{k=1}^{\log n} \frac{n^2}{2^k} = n^2 \sum_{k=1}^b \frac{1}{2^k} = n^2 \sum_{k=1}^b \left(\frac{1}{2} \right)^k = n^2 \sum_{k=1}^b \alpha^k$$

where $\alpha = 1/2$. Let $s_b = \sum_{k=1}^b \alpha^k$. Then:

$$s_b = \alpha + \alpha^2 + \alpha^3 + \dots + \alpha^{b-1} + \alpha^b$$

$$\alpha s_b = \alpha \left(\alpha + \alpha^2 + \alpha^3 + \dots + \alpha^{b-1} + \alpha^b \right) = \alpha^2 + \alpha^3 + \dots + \alpha^b + \alpha^{b+1}$$

$$s_b - \alpha s_b = \alpha + \alpha^2 + \alpha^3 + \dots + \alpha^{b-1} + \alpha^b - \alpha^2 - \alpha^3 - \dots - \alpha^{b-1} - \alpha^b - \alpha^{b+1}$$

$$s_b - \alpha s_b = \alpha - \alpha^{b+1}$$

$$s_b(1 - \alpha) = \alpha(1 - \alpha^b)$$

$$\therefore s_b = \frac{\alpha(1 - \alpha^b)}{1 - \alpha}$$

Using $\alpha = 1/2$:

$$s_b = \frac{\alpha(1 - \alpha^b)}{1 - \alpha} = \frac{(1/2)(1 - (1/2)^b)}{(1 - 1/2)} = \frac{(1/2)\left(1 - \frac{1}{2^b}\right)}{(1/2)}$$

$$\therefore s_b = 1 - 2^{-b}$$

Thus, we can write the function $f_{12}(n)$ as:

$$f_{12}(n) = \sum_{k=1}^{\log n} \frac{n^2}{2^k} = n^2 s_b = (1 - 2^{-b}) n^2$$

Recall $b = \log n$; for $n > 1$, we have that $\log n > 0$, and thus $b > 0$. Therefore, for $n > 1$, $2^{-b} < 1$, and thus the whole prefactor $(1 - 2^{-b}) < 1$. Therefore, for $n > 1$, $(1 - 2^{-b}) n^2 < n^2$.

$$\therefore f_{12}(n) = \sum_{k=1}^{\log n} \frac{n^2}{2^k} \in O(n^2)$$

$$f_{13}(n) = 1 \implies f_{13}(n) \in O(1)$$

$$f_{14}(n) = n^2 + 3n + 5:$$

$$\lim_{n \rightarrow \infty} \frac{f_{14}(n)}{n^2} = \lim_{n \rightarrow \infty} \left(\frac{n^2 + 3n + 5}{n^2} \right) = \lim_{n \rightarrow \infty} \left(1 + \frac{3}{n} + \frac{5}{n^2} \right) = 1$$

$$\therefore f_{14}(n) = n^2 + 3n + 5 \in O(n^2)$$

$$f_{15}(n) = \log n!:$$

We can express $\log n!$ as:

$$\log n! = \log (n \times n!) = \log n + \log ((n-1)!)$$

Using similar logic, we can express $\log ((n-1)!)$ as:

$$\log ((n-1)!) = \log ((n-1) \times (n-2)!) = \log (n-1) + \log ((n-2)!)$$

Substituting above:

$$\log n! = \log n + \log (n-1) + \log ((n-2)!)$$

We can carry out the process for each integer 1 to n , arriving at the sum:

$$\log n! = \sum_{m=1}^n \log m$$

Since $m \leq n$, we can upper-bound the right sum:

$$\begin{aligned} \log n! &= \sum_{m=1}^n \log m \leq \sum_{m=1}^n \log n = \log n \times \sum_{m=1}^n 1 \\ &\therefore \log n! \leq n \log n \quad \forall n \geq 1 \end{aligned}$$

$$\therefore f_{15}(n) = \log n! \in O(n \log n)$$

$$f_{16}(n) = \sum_{k=1}^n \frac{1}{k}:$$

Using equation A.7 in the text:

$$H_n = \sum_{k=1}^n \frac{1}{k} = \ln n + O(1)$$

Thus, taking the limit:

$$\lim_{n \rightarrow \infty} \frac{f_{16}(n)}{\ln n} = \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n \frac{1}{k}}{\ln n} = \lim_{n \rightarrow \infty} \frac{H_n}{\ln n} = \lim_{n \rightarrow \infty} \frac{\ln n + O(1)}{\ln n} = \lim_{n \rightarrow \infty} \left(1 + \frac{O(1)}{\ln n} \right) = 1$$

$$\therefore f_{16}(n) = \sum_{k=1}^n \frac{1}{k} \in O(\ln n)$$

$$f_{17}(n) = \prod_{k=1}^n \left(1 - \frac{1}{k^2} \right):$$

Since the product starts at $k = 1$:

$$\left(1 - \frac{1}{k^2} \right) \Big|_{k=1} = 0$$

Therefore, $\forall n$:

$$\prod_{k=1}^n \left(1 - \frac{1}{k^2} \right) = 0 \leq 1$$

$$\therefore f_{17} = \prod_{k=1}^n \left(1 - \frac{1}{k^2}\right) = 0 < O(1)$$

$$f_{18}(n) = \left(1 - \frac{1}{n}\right)^n:$$

For $n > 1$, we know that $\frac{1}{n} < 1$; thus for $n > 1$, $\left(1 - \frac{1}{n}\right) < 1$.

Therefore:

$$\left(1 - \frac{1}{n}\right)^n \leq 1^n \leq 1 \quad \forall n > 1$$

$$\therefore f_{18}(n) = \left(1 - \frac{1}{n}\right)^n \in O(1)$$

Finally, we can sort the functions in ascending order:

$O(g(n))$	$f(n)$
$T(n) = 0$	$f_{17}(n) = \prod_{k=1}^n \left(1 - \frac{1}{k^2}\right)$
$O(1)$	$f_4(n) = n^{\frac{1}{n}}$
	$f_{13}(n) = 1$
	$f_{18}(n) = \left(1 - \frac{1}{n}\right)^n$
$O(\ln \ln n)$	$f_7(n) = \ln \ln n$
$O(\ln n)$	$f_6(n) = \ln n$
	$f_{16}(n) = \sum_{k=1}^n \frac{1}{k}$
$O(n^\gamma), \gamma = \ln 3$	$f_8(n) = 3^{\ln n}$
$O(n \ln n)$	$f_{15}(n) = \log n!$
$O(n^2)$	$f_1(n) = n^2 + 3n \log n + 5$
	$f_2(n) = n^2 + n^{-2}$
	$f_{11}(n) = n^{1+\cos n}$
	$f_{12}(n) = \sum_{k=1}^{\log n} \frac{n^2}{2^k}$
	$f_{14}(n) = n^2 + 3n + 5$
$O(2^n)$	$f_9(n) = 2^n$
$O(n^n)$	$f_{10}(n) = (1+n)^n$
$O(n^{n^2})$	$f_3(n) = n^{n^2} + n!$
	$f_5(n) = n^{n^2-1}$

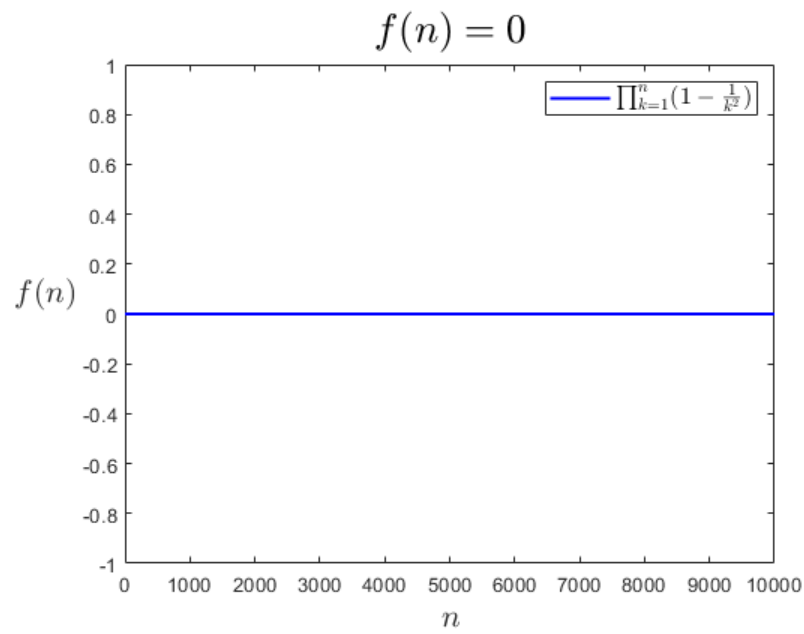


Figure 2: $f(n) = 0$

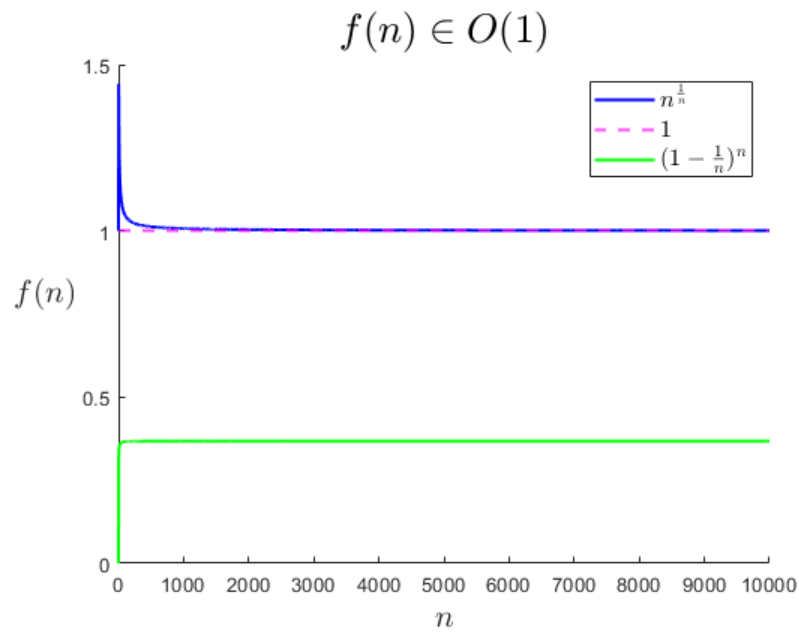


Figure 3: $f(n) \in O(1)$

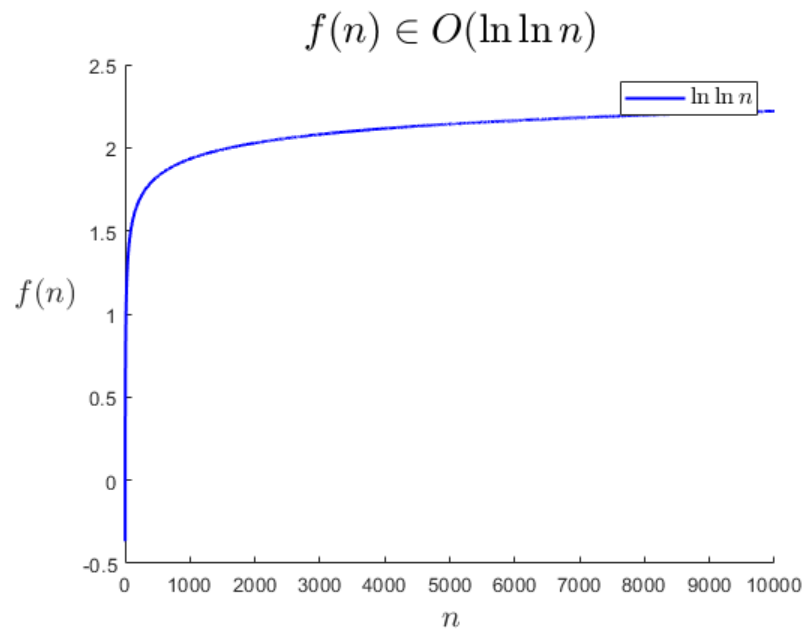


Figure 4: $f(n) \in O(\log \log n)$

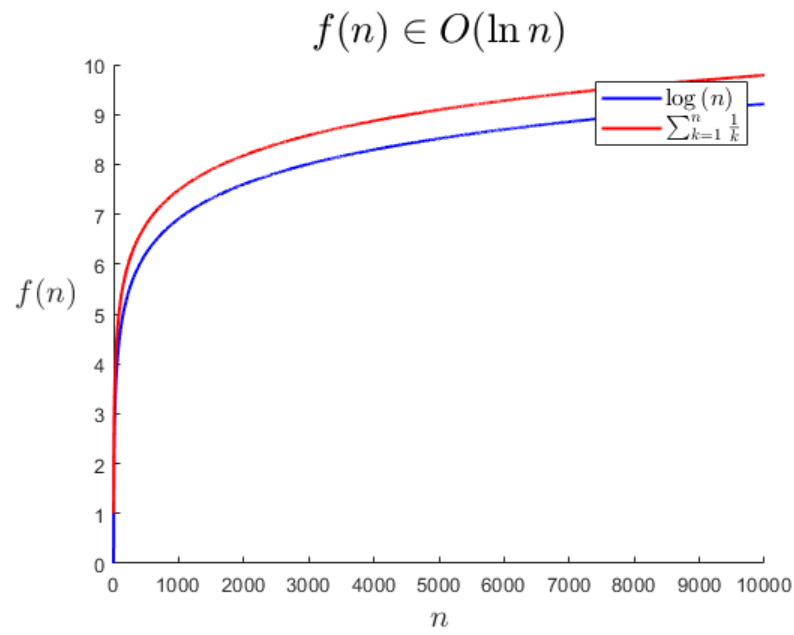


Figure 5: $f(n) \in O(\log n)$

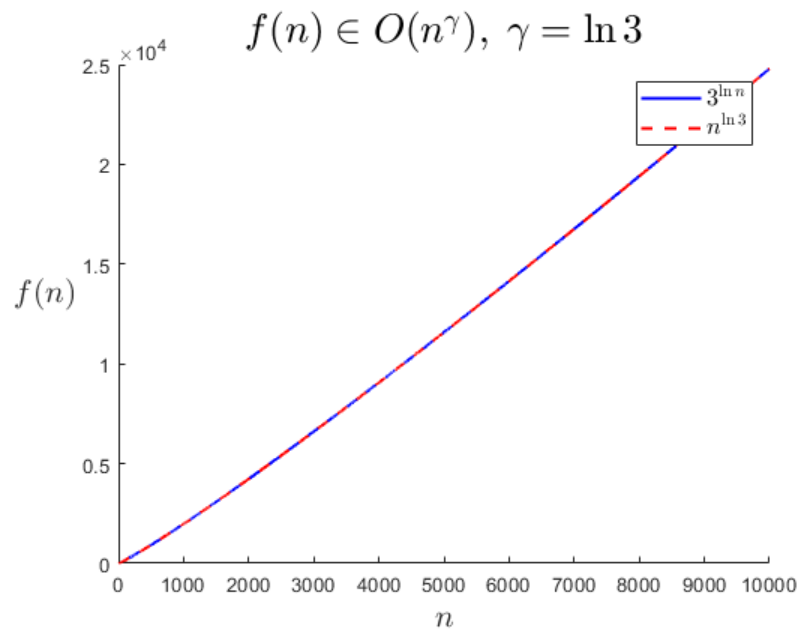


Figure 6: $f(n) \in O(n^\gamma)$

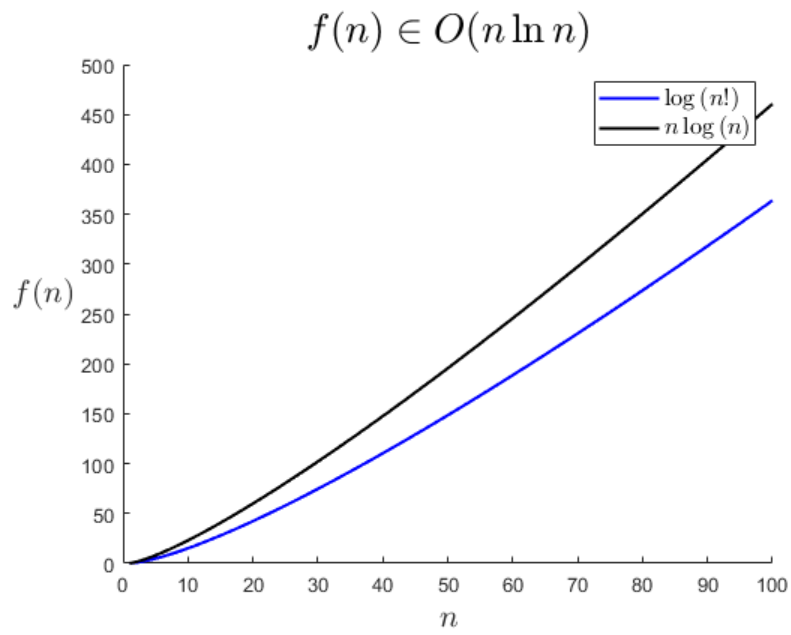


Figure 7: $f(n) \in O(n \log n)$

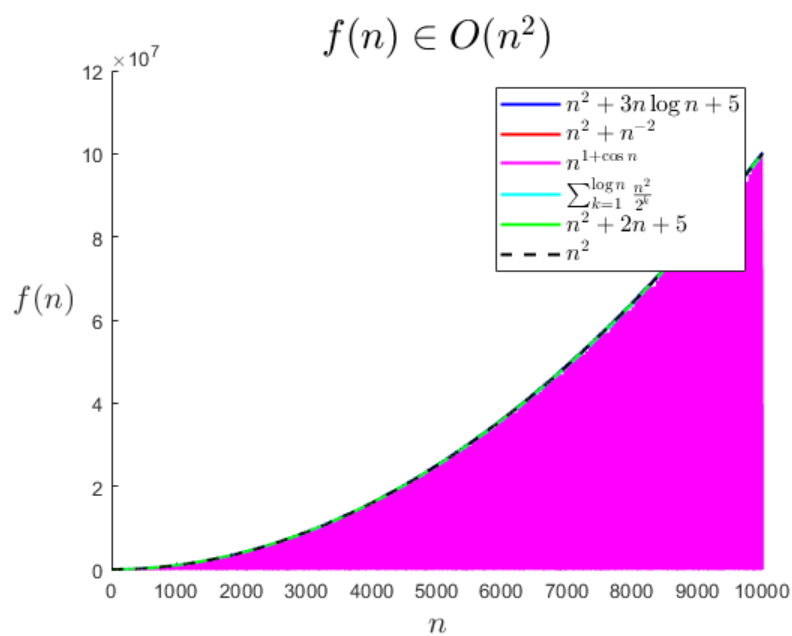


Figure 8: $f(n) \in O(n^2)$

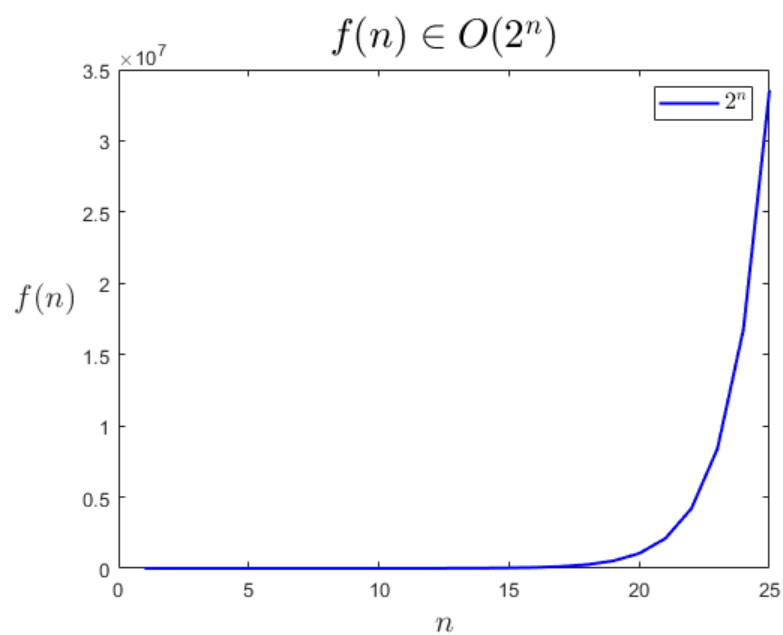


Figure 9: $f(n) \in O(2^n)$

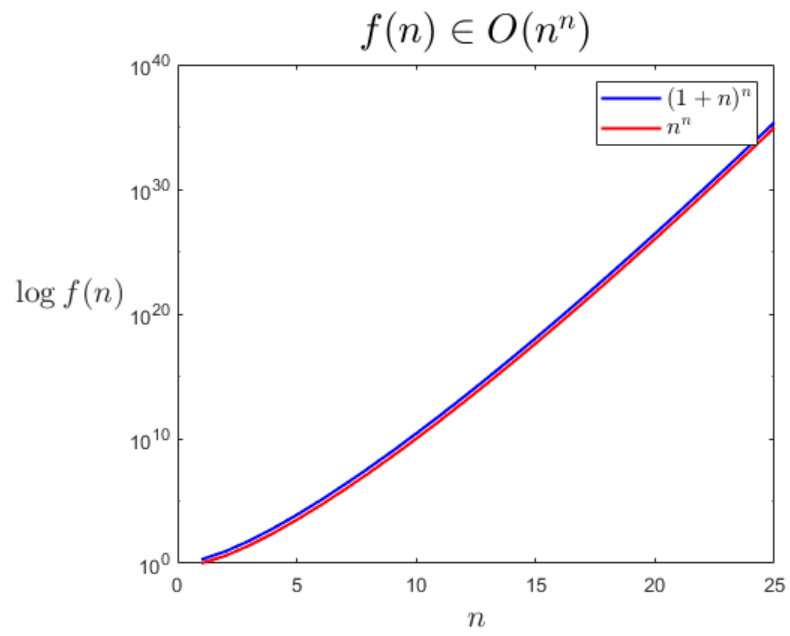


Figure 10: $f(n) \in O(n^n)$

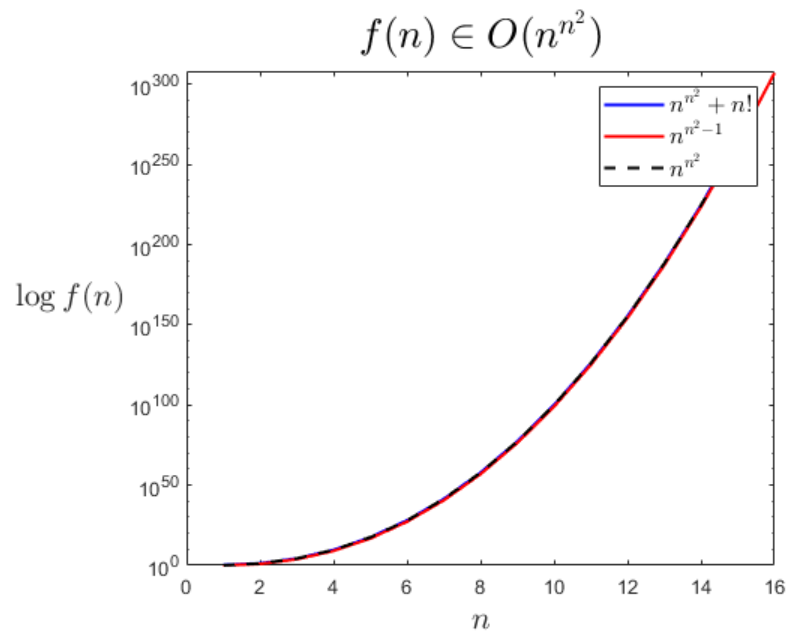


Figure 11: $f(n) \in O(n^{n^2})$

Problem 2

2A

Substitute:

$$T(n) = c_1 n + c_2 n \log_2 n$$

into:

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

to find the values of c_1 , c_2 to determine the exact solution.

$$T(n) = 2T\left(\frac{n}{2}\right) + n \iff c_1 n + c_2 n \log_2 n = 2 \left[c_1 \left(\frac{n}{2}\right) + c_2 \left(\frac{n}{2}\right) \log_2 \left(\frac{n}{2}\right) \right] + n$$

$$c_1 n + c_2 n \log_2 n = c_1 n + c_2 n (\log_2 n - \log_2 2) + n$$

$$c_1 n + c_2 n \log_2 n = c_1 n + c_2 n (\log_2 n - 1) + n$$

$$c_1 n + c_2 n \log_2 n = c_1 n + c_2 n \log_2 n - c_2 n + n$$

$$c_1 n + c_2 n \log_2 n = (c_1 - c_2 + 1)n + c_2 n \log_2 n$$

We require:

$$c_1 = c_1 - c_2 + 1$$

$$\therefore c_2 = 1$$

To find c_1 , assume that $T(1) = t$. Then:

$$T(2) = 2T(1) + 2$$

$$c_1(2) + c_2(2) \log_2(2) = 2t + 2$$

$$2c_1 + 2(1)(1) = 2t + 2 \implies 2c_1 + 2 = 2t + 2$$

$$\therefore c_1 = t = T(1)$$

Therefore, $T(n) = tn + n \log_2 n$ $\square$.

2B

Generalize this to the case for:

$$T(n) = aT\left(\frac{n}{b}\right) + n^k$$

with the trial solution:

$$T(n) = c_1 n^\gamma + c_2 n^k$$

using $a = b^\gamma$.

$$T(n) = aT\left(\frac{n}{b}\right) + n^k \iff c_1 n^\gamma + c_2 n^k = a \left[c_1 \left(\frac{n}{b}\right)^\gamma + c_2 \left(\frac{n}{b}\right)^k \right] + n^k$$

$$c_1 n^\gamma + c_2 n^k = \frac{a c_1 n^\gamma}{b^\gamma} + \frac{a c_2 n^k}{b^k} + n^k$$

Since $a = b^\gamma$:

$$c_1 n^\gamma + c_2 n^k = \frac{b^\gamma c_1 n^\gamma}{b^\gamma} + \frac{b^\gamma c_2 n^k}{b^k} + n^k$$

$$c_1 n^\gamma + c_2 n^k = c_1 n^\gamma + b^{\gamma-k} c_2 n^k + n^k$$

$$c_1 n^\gamma + c_2 n^k = c_1 n^\gamma + (b^{\gamma-k} c_2 + 1) n^k$$

$$c_2 = b^{\gamma-k} c_2 + 1$$

$$c_2 - b^{\gamma-k} c_2 = 1 \implies c_2 (1 - b^{\gamma-k}) = 1$$

$$\therefore c_2 = \frac{1}{1 - b^{\gamma-k}}$$

Again assuming $T(1) = t$, and evaluating at $n = b$:

$$T(b) = aT\left(\frac{b}{b}\right) + b^k$$

$$T(b) = aT(1) + b^k$$

$$c_1 b^\gamma + c_2 b^k = at + b^k$$

$$c_1 b^\gamma = at + b^k - c_2 b^k$$

Using $c_2 = \frac{1}{1 - b^{\gamma-k}}$:

$$c_1 b^\gamma = at + b^k - \left(\frac{1}{1 - b^{\gamma-k}} \right) b^k$$

$$c_1 = \frac{at}{b^\gamma} + \frac{b^k}{b^\gamma} \left(1 - \frac{1}{1 - b^{\gamma-k}} \right)$$

Since $a = b^\gamma$:

$$c_1 = t + \frac{b^k}{b^\gamma} \left(\frac{1 - b^{\gamma-k} - 1}{1 - b^{\gamma-k}} \right)$$

$$c_1 = t + \frac{b^k}{b^\gamma} \left(\frac{-b^{\gamma-k}}{1 - b^{\gamma-k}} \right)$$

$$c_1 = t + \frac{1}{b^\gamma} \left(\frac{-b^\gamma}{1 - b^{\gamma-k}} \right)$$

$$\therefore c_1 = t - \frac{1}{1 - b^{\gamma-k}}$$

Therefore, $T(n) = \left(t - \frac{1}{1-b^{\gamma-k}} \right) n^\gamma + \left(\frac{1}{1-b^{\gamma-k}} \right) n^k \quad \square$.

2C

However, if $\gamma = k$, then the above equation is undefined; thus, the guess $T(n) = c_1 n^\gamma + c_2 n^k$ is no longer valid and we must choose another form. If we guess:

$$T(n) = c_1 n^\gamma + c_2 n^\gamma \log_2 n$$

We can find constants c_1 and c_2 such that the recurrence $T(n) = aT\left(\frac{n}{b}\right) + n^k$ is satisfied for $\gamma = k$.

$$T(n) = aT\left(\frac{n}{b}\right) + n^k \iff c_1 n^\gamma + c_2 n^\gamma \log_2 n = a \left[c_1 \left(\frac{n}{b}\right)^\gamma + c_2 \left(\frac{n}{b}\right)^\gamma \log_2 \left(\frac{n}{b}\right) \right] + n^k$$

$$c_1 n^\gamma + c_2 n^\gamma \log_2 n = \frac{ac_1 n^\gamma}{b^\gamma} + \frac{ac_2 n^\gamma}{b^\gamma} \log_2 \left(\frac{n}{b}\right) + n^k$$

Since $\gamma = k$ and $a = b^\gamma$:

$$c_1 n^\gamma + c_2 n^\gamma \log_2 n = c_1 n^\gamma + c_2 n^\gamma \log_2 \left(\frac{n}{b}\right) + n^\gamma$$

$$c_1 n^\gamma + c_2 n^\gamma \log_2 n = c_1 n^\gamma + c_2 n^\gamma (\log_2 n - \log_2 b) + n^\gamma$$

$$c_1 n^\gamma + c_2 n^\gamma \log_2 n = c_1 n^\gamma + c_2 n^\gamma \log_2 n - (\log_2 b) c_2 n^\gamma + n^\gamma$$

$$c_1 n^\gamma + c_2 n^\gamma \log_2 n = c_2 n^\gamma \log_2 n + (c_1 + 1 - c_2 \log_2 b) n^\gamma$$

$$c_1 = c_1 + 1 - c_2 \log_2 b$$

$$\therefore c_2 = \frac{1}{\log_2 b}$$

Again assuming $T(1) = t$ and evaluating at $n = b$:

$$T(b) = aT\left(\frac{b}{b}\right) + b^\gamma$$

$$T(b) = aT(1) + b^\gamma$$

$$c_1 b^\gamma + c_2 b^\gamma \log_2 b = at + b^\gamma$$

$$c_1 b^\gamma = at + b^\gamma - c_2 b^\gamma \log_2 b$$

$$c_1 = \frac{at}{b^\gamma} + \frac{b^\gamma}{b^\gamma} (1 - c_2 \log_2 b)$$

Using $a = b^\gamma$ and $c_2 = \frac{1}{\log_2 b}$:

$$c_1 = t + \left(1 - \frac{\log_2 b}{\log_2 b}\right)$$

$$\therefore c_1 = t$$

Therefore, $T(n) = tn^\gamma + \left(\frac{\log_2 n}{\log_2 b}\right)n^\gamma \quad \square$.

(30 pts) Place the following functions in order from asymptotically smallest to largest. As a convenience you may use $f(n) < O(g(n))$ to mean $f(n) \in O(g(n))$ and $f(n) = g(n)$ to mean $f(n) \in \Theta(g(n))$. Please use $=$ when you are sure that it is $\Theta(g(n))$.)

$$n^2 + 3n \log(n) + 5, \quad n^2 + n^{-2}, \quad n^{n^2} + n!, \quad n^{\frac{1}{n}}, \quad n^{n^2-1}, \quad \ln n, \quad \ln(\ln n), \quad 3^{\ln n}, \quad 2^n, \\ (1+n)^n, \quad n^{1+\cos n}, \quad \sum_{k=1}^{\log n} \frac{n^2}{2^k}, \quad 1, \quad n^2 + 3n + 5, \quad \log(n!), \quad \sum_{k=1}^n \frac{1}{k}, \quad \prod_{k=1}^n (1 - \frac{1}{k^2}), \quad (1 - 1/n)^n$$

Giving the the algebra and explanation for the tricky cases can get some extra credit (even if you get it wrong!). Don't have to be perfect to get a good score.

Solution: SOME OF THESE ARE PRETTY SUBTLE. A GOOD RESULT IS GETTING THE ORDER SOME WHAT RIGHT. Here I give much more details not required since they are instructive. In some cases I even give the exact limit as $\rightarrow xx$.

$$1 \in O(1) \quad , \quad \prod_{k=1}^n (1 - \frac{1}{k^2}) \rightarrow 0 \in O(1) \quad , \quad n^{\frac{1}{n}} \in O(1) \quad , \quad (1 - 1/n)^n \rightarrow e \in O(1)$$

$$\ln(\ln n) \in O(\ln(\ln n)) \quad , \quad \ln(n) \in O(\log(n)) \quad , \quad \sum_{k=1}^n \frac{1}{k} = O(\ln n)$$

$$\ln(n!) \in O(n \log(n)) \quad , \quad 3^{\ln n} \rightarrow n^{\ln(3)} \in O(n^{\ln(3)}) \quad , \quad n^{1+\cos n} \in O(n^2) \\ n^2 + 3n + 5 \in O(n^2) \quad , \quad n^2 + 3n \ln n + 5 \in O(n^2) \quad , \quad n^2 + n^{-2} \in O(n^2)$$

$$2^n \in O(2^n) \quad , \quad \sum_{k=1}^{\log n} \frac{n^2}{2^k} \in ? \quad , \quad n! \in O(e^{-n} n^{n+1/2}) \quad , \quad (1+n)^n \in O(n^n)$$

$$n^{n^2-1} \in O(\frac{1}{n} n^{n^2}) \quad , \quad n^{n^2} + n! \in O(n^{n^2})$$

Here are few explicit sums:

1:

For instance,

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n (1 - \frac{1}{k^2}) = 0 \in O(0)$$

because the first term in the product is zero. If the first term were not 0, then the product would converge to a constant, as

$$\ln \prod_{k=2}^n (1 - \frac{1}{k^2}) = \sum_{k=1}^n \ln(1 - \frac{1}{k^2}) \approx \sum_{k=1}^n -\frac{1}{k^2} = \frac{-\pi^2}{6}$$

2:

$\sum_{k=1}^{\ln n} \frac{n^2}{2^k} = n^2 \sum_{k=1}^{\ln n} (\frac{1}{2})^k = O(n^2)$. Just realizing that a geometric series is convergent $\sum_{k=0}^N x^k = O(1)$ or if you believe in slugging it through (not necessary!). $\sum_{k=0}^N x^k = \frac{1-x^{N+1}}{1-x}$ so $\sum_{k=1}^N x^k = \frac{1-x^{N+1}}{1-x} - 1 = \frac{1-(1/2)^{\ln(n)+1}}{1/2} - 1 = 1 - (1/2)^{\ln(n)} \in O(1)$ using $x = 1/2$ and $N = \ln(n)$.

3:

In general a very useful trick is to take **ln-exp!** of the function $(f(n) = e^{\ln(f(n))})$ followed by the large n limit. For example:

$$(1 - 1/n)^n = e^{n \ln(1-1/n)} \simeq e^{n(1/n + 1/2n^2 + \dots)} \rightarrow e^1 = 2.718281828459$$

(I found this going to WolframAlpha: <https://www.wolframalpha.com!>) Also, recall that $e^{\ln n} = n$.

4:

Finally the function $n^{1+\cos n}$ is tricky so we accept any reasonable placements. It doesn't have a smooth monotonic limit at large n . It oscillates between $\Theta(1)$ and $\Theta(n^2)$ getting arbitrarily close both even at integer values. Therefore strictly speaking the best bound is $n^{1+\cos n} \in O(n^2)$ but $n^\alpha \in O(n^{1+\cos n})$ implies $\alpha \leq 0$ by the definition in CLRS of "Big Oh".

1. (40 pts)

- (a) Given the equation, $T(n) = 2T(n/2) + n$, guess a solution of the form:

$$T(n) = c_1 n + c_2 n \log_2(n) .$$

Find the coefficients c_1, c_2 to determine the exact solution assuming a value $T(1)$ at the bottom the recursion.

- (b) Generalize this to the case to the equation $T(n) = aT(n/b) + n^k$ and guess the solution of the form:

$$T(n) = c_1 n^\gamma + c_2 n^k$$

using $b^\gamma = a$ and assuming $\gamma \neq k$. First show if you drop the n^k using the **homogeneous** equation $T(n) = aT(n/b)$ the form $c_1 n^\gamma$ is a solution! (What is c_1 ?) Second drop $aT(n/b)$ and show $c_2 n^k$ is a solution (What is c_2 ?) With both terms (and $\gamma \neq k$) the full solution is just the sum of the two terms but only one or the other dominates!

- (c) What happens when the two solutions collide (i.e. have the same power, i.e. $\gamma = \log(a)/\log(b) = k$.) Now show that the leading solution is as n goes to infinity is $T(n) = \Theta(n^k \log n)$ ¹

¹If you are ambitious you find exact solutions for part b and c above are of the form $T(n) = c_1 n^\gamma + \text{give } c_2 n^k$ and $T(n) = c_1 n^k + c_2 n \log(n) n^k$ explicitly determine the c 's for each case respectively. We did that in part for example. BUT note we already know the leading terms for the Master Equation without this extra effort!
Neat trick!

Solution: You are given a **guess** with unknown constants c_1, c_2 . To see if a good guess see if you can determine the constants by substituting the **guess** in the RHS (right hand side) and the LHS (left hand side).

First case:

$$\begin{aligned} c_1 n + c_2 n \log_2(n) &= 2[c_1 n/2 + c_2 (n/2) \log_2(n/2)] + n \\ &= c_1 n + c_2 n (\log_2(n) - \log_2(2)) + n \\ &= c_1 n + c_2 n \log_2(n) - n c_2 + n \end{aligned}$$

Now to see if $\text{RHS} = \text{LHS}$ $c_1 = c_2$ you need to match the n and the $n \log_2(n)$ term

$$c_1 = c_1 - c_2 + 1 \quad \text{and} \quad c_2 = c_2 \quad (1)$$

so it works $c_2 = 1$ and any c_1 . To determine this you need to base provide a base case of the recursion (e.g. $T(1) = c_1$).

Second case: With $\gamma \neq k$ the general solution works. Again LHS vs RHS

$$c_1 n^\gamma + c_2 n^k = a[c_1 (n/b)^\gamma + c_2 (n/b)^k] + n^k \quad (2)$$

Two conditions to match term:

$$c_1 = c_1 a/b^\gamma \quad \text{and} \quad c_2 = a/b^k c_2 + 1 \quad (3)$$

Again it works for any c_1 but we need $c_2 = 1/(1 - a/b^k)$. Trouble if $k = \gamma$ because now $c_2 = \infty$ (i.e. it fails!)

So need to start over with something larger by a log. Again try to match LHS and RHS

$$c_1 n^\gamma + c_2 n^\gamma \log_2(n) = a[c_1 (n/b)^\gamma + c_2 (n/b)^\gamma \log_2(n/b)] + n^\gamma \quad (4)$$

Now the leading term c_2 works and the lower power is determined

$$c_2 = c_2 \quad \text{and} \quad c_1 = c_1 - c_2 \log_2(b) \quad (5)$$

lower term is again determined relative to it. This is general the larger term matches as n goes to infinity and the smaller needs a base number of $T(1)$.